

1. **Data mining:**

Data mining is the process of finding useful patterns and knowledge from large datasets.

Two tasks: 1. Classification
2. Clustering

KDD steps: Selection, cleaning, Integration, Transformation, Data mining, pattern Evaluation, knowledge presentation

Role of preprocessing: 1. Improves data quality
2. Removes errors
3. Increases accuracy

b)

Data cleaning: Removing incorrect or incomplete data

Data quality issues: 1. missing values
2. Noisy data

* missing values: Replace with mean/median/mode or remove records

* Noisy data: use binning, regression, or clustering

Data integration:

Data integration is the process of combining data from multiple sources into a single, consistent dataset.

Two challenges:

1. Data Redundancy (Duplicate data)

2. Data Inconsistency (different data formats)

→ Inconsistency; Standardization

(b) Covariance Analysis:
Covariance analysis measures how two variables move together.

Two Covariance Formulas:

1. Population Covariance:
$$\text{Cov}(X, Y) = \frac{\sum (x - \mu_x)(y - \mu_y)}{N}$$
2. Sample Covariance:
$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{n - 1}$$

Interpretation:

Positive Covariance: variables increase or decrease together
Negative Covariance: one increases while the other decreases.

Given data

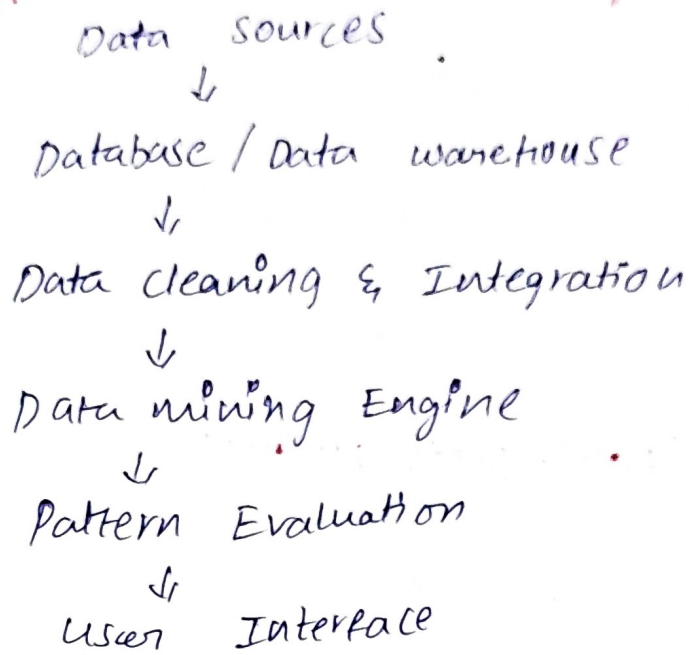
Stock A	Stock B
2	5
3	8
5	10
4	11
6	14

Answer:

As the values of Stock A increase, the values of Stock B also increase. Therefore, the stocks have positive covariance (positive correlation).

Conclusion: The stock prices rise together and are not independent.

Architecture of a data Mining System



Components:

Data Sources: Collect raw data

Database / Data warehouse: Store integrated data.

Data cleaning & Integration: Improved data quality

Data mining Engine: Finds patterns using algorithms

Pattern Evaluation: Select useful patterns

User Interface: Displays the results to users.

4. Given:

$$X = \{4, 5, 6, 7, 9\}$$

$$Y = \{8, 3, 7, 5, 6\}$$

$$\text{Mean}(X) = 6.2$$

$$\text{Mean}(Y) = 5.8$$

Using the covariance formula,
$$\text{Cov}(X, Y) = \frac{\sum (x - \bar{x})(y - \bar{y})}{N}$$

$$\text{Covariance} = -0.36$$

$$\text{Answer: } \text{Cov}(X, Y) = -0.36$$

(b) major steps in data preprocessing

Data cleaning, Data Integration, Data transformation, Data reduction, Data discretization

preprocessing pipeline: collect data

↓
data cleaning → handling values

↓
data mining ← Normalize data ← Remove outliers

5) Equal Frequency (Equal-Depth) Binning:

Total values = 27, Number of bins = 3, values per bin = 9

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22 → Mean = 18

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35 → Mean = 28

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70 → Mean = 44

Answer: smoothed values are 18, 28 and 44 for the three bins

(b) Data discretization

Converts continuous data into intervals (bins)

Importance

- Reduce data complexity
- Improve analysis
- Remove noise

Comparison

Equal-width

Same bin width

Simple

→ Equal Frequency

→ Same number of values

→ Balanced data

Impact: Equal Frequency is better for uneven data, while equal width is suitable for uniform data.